

A general method for constructing quasi-interpolants from B-splines[☆]

A. Abbadi^a, D. Barrera^b, M.J. Ibáñez^{b,*}, D. Sbibi^a

^a Département de Mathématiques et Informatique, Université Mohammed 1er, 60000-Oujda, Maroc

^b Departamento de Matemática Aplicada, Universidad de Granada, 18071-Granada, Spain

ARTICLE INFO

Article history:

Received 1 October 2008

Received in revised form 18 January 2010

Keywords:

B-splines

Box splines

Differential quasi-interpolants

Discrete quasi-interpolants

Integral quasi-interpolants

Error estimates

Approximation power

ABSTRACT

A general method for constructing quasi-interpolation operators based on B-splines is developed. Given a B-spline ϕ in $\mathbb{R}^s$, $s \geq 1$, normalized by $\sum_{i \in \mathbb{Z}^s} \phi(\cdot - i) = 1$, the classical structure $Q(f) := \sum_{i \in \mathbb{Z}^s} \lambda f(\cdot + i) \phi(\cdot - i)$, for a quasi-interpolation operator Q is considered. A minimization problem is derived from an estimate of the quasi-interpolation error associated with Q when λf is a linear combination of values of f at points in some neighbourhood of the support of ϕ ; or a linear combination of values of f and some of its derivatives at some points in this set; or a linear combination of weighted mean values of the function to be approximated. That linear functional is defined to produce a quasi-interpolant exact on the space of polynomials of maximal total degree included in the space spanned by the integer translates of ϕ . The solution of that minimization problem is characterized in terms of specific splines which do not depend on λ but only on ϕ .

© 2010 Elsevier B.V. All rights reserved.

1. Introduction and notations

We propose a general method for constructing quasi-interpolation operators based on a B-spline, i.e. a compactly supported non-negative polynomial piecewise function defined on $\mathbb{R}^s$, $s \geq 1$. For such a function ϕ , normalized by $\sum_{i \in \mathbb{Z}^s} \phi(\cdot - i) = 1$, let $\mathcal{S}(\phi) := \text{span}(\phi(\cdot - i))_{i \in \mathbb{Z}^s}$ be the cardinal spline space spanned by the shifts of ϕ . As quoted in [1, pp. 63], a quasi-interpolant on $\mathcal{S}(\phi)$ is a linear map into $\mathcal{S}(\phi)$ which is local, bounded, and reproduces some (nontrivial) polynomial space. The classical structure for a quasi-interpolant is given by the expression

$$Qf := Q(f) := \sum_{i \in \mathbb{Z}^s} \lambda f(\cdot + i) \phi(\cdot - i), \quad (1)$$

λ being some suitable linear functional.

In this paper, we consider the linear functional λ given by one of the following three expressions:

$$\lambda f = \sum_{j \in J} c_j f(-j), \quad (2)$$

$$\lambda f = \sum_{|\alpha| \leq \ell} \sum_{j \in J} c_{\alpha,j} \partial^\alpha f(-j), \quad (3)$$

$$\lambda f = \sum_{j \in J} c_j \langle f, \psi(\cdot - j) \rangle, \quad (4)$$

[☆] Research supported in part by Junta de Andalucía (Research group FQM/191) and by the Ministerio de Ciencia e Innovación of Spain (Research project MTM2008-00671).

* Corresponding author.

E-mail addresses: abzabbadi@yahoo.fr (A. Abbadi), dbarrera@ugr.es (D. Barrera), mibanez@ugr.es (M.J. Ibáñez), sbibi@yahoo.fr (D. Sbibi).

J and J_α , $|\alpha| \leq \ell$ for $0 \leq \ell \leq \deg \phi$, being finite subsets of $\mathbb{Z}^s$, $\langle \cdot, \cdot \rangle$ standing for the usual inner product and ψ some other B-spline such that $\int_{\mathbb{R}^s} \psi = 1$. We will refer to discrete, differential or integral quasi-interpolants, respectively. By c we denote the sequence $(c_j)_{j \in J}$ defining Q in the discrete and integral cases, and $(c_{\alpha,j})_{|\alpha| \leq \ell, j \in J_\alpha}$ in the differential one. In general, we omit any reference to ℓ, J and J_α in writing c .

In addition to the standard procedures that appear in the literature for constructing quasi-interpolants of these types (see [1, pp. 68–78], [2, pp. 121–128], [3–7] and [8, pp. 359–363], and [9,10] in the bivariate case); other methods have been developed to define quasi-interpolants with specific properties.

In [11–15] the construction of near-best discrete and integral quasi-interpolants is addressed, i.e. quasi-interpolation operators having small infinity norms (see [16] in the non-uniform case, [17] in relation to a similar minimization problem defined from the Bernstein–Bézier representation of the B-spline, and [18] for a non-standard method). Taking into account the error estimate $\|f - Qf\|_\infty \leq (1 + \|Q\|_\infty) \text{dist}(f, \mathcal{R})$ for a given quasi-interpolation operator reproducing some space $\mathcal{R}$, the near-best quasi-interpolants contribute to obtain small constants in the error estimates. Nevertheless, it is possible to construct discrete and integral quasi-interpolants by examining closely the associated errors for regular enough functions. With respect to the univariate discrete case, and using odd order B-splines, a construction based on the minimization of a term in an appropriate error estimate is given in [19]. In that construction, the symmetry of the sequence c defining the operator is imposed. The method is extended in [20] to consider any B-spline and any sequence c . The bivariate case appears in [17], where some remarkable low degree discrete quasi-interpolants are obtained. The investigation of the integral case starts in [21] and is extended in [22].

In each of these contributions, the solutions of the different minimization problems have characterizations depending on the specific form of the linear functional λ . In this paper, we propose a general construction based on some splines which do not depend on λ but only on the B-spline ϕ .

In general, λ is defined to produce an operator Q exact on a polynomial space included in $\mathcal{S}(\phi)$. More precisely, we will demand the exactness of Q on $\mathbb{P}_n$, with n such that $\mathbb{P}_n \subset \mathcal{S}(\phi)$ and $\mathbb{P}_{n+1} \not\subset \mathcal{S}(\phi)$, i.e. Q realizes the approximation power of $\mathcal{S}(\phi)$. We will restrict our attention to the three linear functionals in (2)–(4).

The paper is organized as follows. In Section 2, an appropriate estimate for the quasi-interpolation error is given. In Section 3, a minimization problem is proposed and its solutions are characterized. In Section 4, some C^2 quartic quasi-interpolants are derived. In Section 5, the differential schemes are tested and compared with some other operators.

Throughout this paper, we use the following notations. For a multi-index $\alpha := (\alpha_1, \dots, \alpha_s) \in \mathbb{R}^s$ with non-negative components, we write $|\alpha| := \sum_{k=1}^s \alpha_k$. The factorial of α is given by $\alpha! := \prod_{k=1}^s \alpha_k!$, and the multivariate monomial of order α is defined by $x^\alpha := \prod_{k=1}^s x_k^{\alpha_k}$. Moreover, we use the notations $m_\alpha(x) := x^\alpha / \alpha!$ for the normalized monomial of order α , and $\partial^\alpha u(x)$ for the partial derivative of order α of u at x . We note $\|u\|_{\infty, S} := \sup_{x \in S} |u(x)|$ for the max-norm of u on a compact set $S \subset \mathbb{R}^s$, and $|u|_{\infty, m, S} := \sum_{|\alpha|=m} \|\partial^\alpha u\|_{\infty, S}$ for the semi-norm of order $m \in \mathbb{N}$.

2. Quasi-interpolation error

The starting point of the construction we propose is a pointwise estimate of $E_h f := f - Q_h f$ for the scaled quasi-interpolant

$$Q_h f := \sum_{i \in \mathbb{Z}^s} \lambda f((\cdot + i)h) \phi\left(\frac{\cdot}{h} - i\right). \quad (5)$$

Proposition 1. Let Q_h be a quasi-interpolation operator defined by (5) and one of the linear functionals (2), (3) or (4), and suppose Q is exact on $\mathbb{P}_n$. Let $f \in C^{n+2}(\mathbb{R}^s)$. Then, for every $x \in \mathbb{R}^s$ there exist both a neighbourhood V of x , independent of h , and a constant $C > 0$, independent of h and x , such that

$$|E_h f(x)| \leq F_n(c) h^{n+1} \sum_{|\alpha|=n+1} |\partial^\alpha f(x)| + C h^{n+2} |f|_{\infty, n+2, V},$$

where

$$F_n(c) := F_{n, \phi, \psi}(c) := \max_{|\alpha|=n+1} \|Qm_\alpha - m_\alpha\|_{\infty, [0,1]^s}.$$

Proof. Let Q be the operator defined by (5) and (2). Taking the Taylor expansion

$$f(y) = \sum_{|\alpha| \leq n+1} \frac{(y-x)^\alpha}{\alpha!} \partial^\alpha f(x) + \sum_{|\alpha|=n+2} \frac{(y-x)^\alpha}{\alpha!} \partial^\alpha f(\tau x + (1-\tau)y), \quad \tau \in [0, 1],$$

for each $f(h(i-j))$, we write the discrete quasi-interpolant $Q_h f$ in the form

$$Q_h f(x) = \mathcal{A}_h(x) + \mathcal{B}_h(x),$$

where

$$\mathcal{A}_h(x) := \sum_{i \in \mathbb{Z}^s} \left(\sum_{j \in J} c_j \sum_{|\alpha| \leq n+1} \frac{1}{\alpha!} \partial^\alpha f(x) \frac{(h(i-j) - x)^\alpha}{\alpha!} \right) \phi\left(\frac{x}{h} - i\right),$$

and

$$\mathcal{B}_h(x) := \sum_{i \in \mathbb{Z}^s} \left(\sum_{j \in J} c_j \sum_{|\alpha| = n+2} \partial^\alpha f(\tau_i x + (1 - \tau_i) h(i-j)) \right) \phi\left(\frac{x}{h} - i\right),$$

with $\tau_i \in [0, 1]$ for all $i \in \mathbb{Z}^s$.

Since

$$\begin{aligned} \mathcal{A}_h(x) &= f(x) + \sum_{1 \leq |\alpha| \leq n+1} \frac{1}{\alpha!} \partial^\alpha f(x) \sum_{i \in \mathbb{Z}^s} \left(\sum_{j \in J} c_j ((i-j)h - x)^\alpha \right) \phi\left(\frac{x}{h} - i\right) \\ &= f(x) + \sum_{1 \leq |\alpha| \leq n+1} \frac{1}{\alpha!} \partial^\alpha f(x) Q_h[(\cdot - x)^\alpha](x) \end{aligned}$$

and Q_h is exact on $\mathbb{P}_n$, we obtain

$$\mathcal{A}_h(x) = f(x) + \sum_{|\alpha| = n+1} \frac{1}{\alpha!} \partial^\alpha f(x) Q_h[(\cdot - x)^\alpha](x).$$

Therefore

$$|Q_h f(x) - f(x)| \leq \sum_{|\alpha| = n+1} |\partial^\alpha f(x)| \left| Q_h \left[\frac{(\cdot - x)^\alpha}{\alpha!} \right](x) \right| + |\mathcal{B}_h(x)|.$$

Taking into account that

$$q_{\alpha,h}(x) := \left| Q_h \left[\frac{(\cdot - x)^\alpha}{\alpha!} \right](x) \right|$$

satisfies the periodicity condition

$$q_{\alpha,h}(x + h\beta) = q_{\alpha,h}(x), \quad \beta \in \mathbb{Z}^s,$$

we deduce that

$$\begin{aligned} |Q_h f(x) - f(x)| &\leq \sum_{|\alpha| = n+1} |\partial^\alpha f(x)| \left\| Q_h \left[\frac{(\cdot - x)^\alpha}{\alpha!} \right] \right\|_{\infty, [0,h]^s} + |\mathcal{B}_h(x)| \\ &= h^{n+1} \sum_{|\alpha| = n+1} |\partial^\alpha f(x)| \left\| Q \left[\frac{(\cdot - \xi)^\alpha}{\alpha!} \right] \right\|_{\infty, [0,1]^s} + |\mathcal{B}_h(x)| \\ &\leq h^{n+1} \max_{|\alpha| = n+1} \left\| Q \left[\frac{(\cdot - \xi)^\alpha}{\alpha!} \right] \right\|_{\infty, [0,1]^s} \sum_{|\alpha| = n+1} |\partial^\alpha f(x)| + |\mathcal{B}_h(x)|. \end{aligned}$$

By the exactness of Q on $\mathbb{P}_n$, we have

$$\max_{|\alpha| = n+1} \left\| Q \left[\frac{(\cdot - \xi)^\alpha}{\alpha!} \right] \right\|_{\infty, [0,1]^s} = F_n(c),$$

and then it only remains to estimate $|\mathcal{B}_h(x)|$.

Note that, for a fixed $x \in \mathbb{R}^s$, only finite many indices i are involved in the sum defining $\mathcal{B}_h(x)$ because ϕ is a compactly supported function. Moreover, J is a finite subset. After some calculations, we obtain

$$|\mathcal{B}_h(x)| \leq h^{n+2} \sum_{|\alpha| = n+2} \|\partial^\alpha f\|_{\infty, V} W_{\alpha,h}(x),$$

where V is some closed ball centered at x containing all segments with extreme points x and $h(i-j)$, and

$$W_{\alpha,h}(x) := \sum_{i \in \mathbb{Z}^s} \left(\sum_{j \in J} |c_j| \left| \frac{(i-j-\frac{x}{h})^\alpha}{\alpha!} \right| \right) \phi\left(\frac{x}{h} - i\right).$$

Since $W_{\alpha,h}(x + \beta h) = W_{\alpha,h}(x)$, $\beta \in \mathbb{Z}^s$, we get

$$\begin{aligned}
|\mathcal{B}_h(x)| &\leq h^{n+2} \sum_{|\alpha|=n+2} \|\partial^\alpha f\|_{\infty,V} \|W_{\alpha,h}\|_{\infty,[0,h]^s} \\
&\leq h^{n+2} \max_{|\alpha|=n+2} \|W_{\alpha,h}\|_{\infty,[0,h]^s} \sum_{|\alpha|=n+2} \|\partial^\alpha f\|_{\infty,V} \\
&= h^{n+2} \max_{|\alpha|=n+2} \|W_{\alpha,1}\|_{\infty,[0,1]^s} \|f\|_{\infty,n+2,V},
\end{aligned}$$

and the proof is complete taking $C = \max_{|\alpha|=n+2} \|W_{\alpha,1}\|_{\infty,[0,1]^s}$.

When Q is the operator defined by (5) and (4) a similar proof holds, but now

$$\mathcal{A}_h(x) := \sum_{i \in \mathbb{Z}^s} \left(\sum_{j \in J} c_j \sum_{|\alpha| \leq n+1} \frac{1}{\alpha!} \partial^\alpha f(x) < (h(\cdot + i) - x)^\alpha, \psi(\cdot - j) > \right) \phi\left(\frac{x}{h} - i\right),$$

and

$$\mathcal{B}_h(x) := \sum_{i \in \mathbb{Z}^s} \left(\sum_{j \in J} c_j \sum_{|\alpha|=n+2} \int_{\mathbb{R}^s} \frac{(h(t+i) - x)^\alpha}{\alpha!} \partial^\alpha f(\pi_{i,h}(x, t)) \psi(t - j) dt \right) \phi\left(\frac{x}{h} - i\right),$$

with

$$\pi_{i,h}(x, t) := \tau_i x + (1 - \tau_i) h(t + i),$$

and $\tau_i \in [0, 1]$ for all $i \in \mathbb{Z}^s$. Once again

$$\mathcal{A}_h(x) = f(x) + \sum_{|\alpha|=n+1} \frac{1}{\alpha!} \partial^\alpha f(x) Q_h[(\cdot - x)^\alpha](x),$$

and then

$$|Q_h f(x) - f(x)| \leq \sum_{|\alpha|=n+1} |\partial^\alpha f(x)| \left| Q_h \left[\frac{(\cdot - x)^\alpha}{\alpha!} \right] (x) \right| + |\mathcal{B}_h(x)|,$$

where

$$|\mathcal{B}_h(x)| \leq h^{n+2} \sum_{|\alpha|=n+2} \|\partial^\alpha f\|_{\infty,V} W_{\alpha,h}(x),$$

V being some closed ball centered at x containing all segments with extreme points x and $h(t + i)$, and

$$W_{\alpha,h}(x) := \sum_{i \in \mathbb{Z}^s} \left(\sum_{j \in J} |c_j| \int_{\mathbb{R}^s} \left| \frac{(t + i - \frac{x}{h})^\alpha}{\alpha!} \right| \psi(t - j) dt \right) \phi\left(\frac{x}{h} - i\right).$$

Also in this case, the claim follows with $C = \max_{|\alpha|=n+2} \|W_{\alpha,1}\|_{\infty,[0,1]^s}$ taking into account that $W_{\alpha,h}(x + \beta h) = W_{\alpha,h}(x)$, $\beta \in \mathbb{Z}^s$. For a differential quasi-interpolation operator the proof is very similar. $\square$

The constant $F_n(c)$ in the leading term of the pointwise estimate for $E_h f$ in Proposition 1 is determined by how well Q_h approximates the monomials of total degree $n + 1$ (see [23]).

The reproduction of $\mathbb{P}_n$ by Q is achieved if λ agrees on $\mathbb{P}_n$ with the linear functional $\lambda_0 : f \mapsto \sum_{\alpha} g_{\alpha}(0) \partial^\alpha f(0)$, $(g_{\alpha})_{\alpha \in \mathbb{N}_0^s}$ being the Appell sequence for the linear functional μ given by $\mu f := \sum_{j \in \mathbb{Z}^s} f(j) \phi(-j)$, i.e. $\mu \partial^\beta g_{\alpha} = \delta_{\beta,\alpha}$ where $\delta_{\beta,\alpha}$ denotes the Kronecker delta (see [1, p. 68] and the discussion on the construction of g_{α}). Therefore, Q is exact on $\mathbb{P}_n$ if for all $\alpha \in \mathbb{N}_0^s$ such that $|\alpha| \leq n$ one gets

$$\lambda(m_{\alpha}) = g_{\alpha}(0), \quad (6)$$

where the polynomials g_{α} can be recursively computed as follows:

$$g_0 = m_0, \quad g_{\alpha} = m_{\alpha} - \sum_{j \in \mathbb{Z}^s} \phi(j) \sum_{\beta \leq \alpha} m_{\alpha-\beta}(-j) g_{\beta}, \quad |\alpha| > 0. \quad (7)$$

They are only sufficient conditions to guarantee the exactness of Q on $\mathbb{P}_n$.

3. A minimization problem

The two-term pointwise estimate in Proposition 1 suggests to construct the operator Q minimizing the function $F_n(c)$ in the leading term of that estimate. The exactness conditions need to be imposed. So, we propose

Problem 2. Minimize $F_n(c)$ under the constraints (6).

This problem can be easily solved and its solutions (and the corresponding quasi-interpolation operators Q) can be easily characterized using the well-known Schoenberg operator

$$Sf := \sum_{i \in \mathbb{Z}^s} f(i) \phi(\cdot - i). \quad (8)$$

Proposition 3. Let Q be the quasi-interpolation operator associated with one of the linear functionals λ defined by (2)–(4). Let us suppose that (6) holds. If the conditions

$$\lambda m_\alpha = g_\alpha(0) + \frac{1}{2} \left(\max_{[0,1]^s} G_\alpha + \min_{[0,1]^s} G_\alpha \right) \quad (9)$$

hold for all $\alpha \in \mathbb{N}_0^s$ such that $|\alpha| = n + 1$, where

$$G_\alpha := m_\alpha - Sg_\alpha, \quad (10)$$

then $F_n(c)$ attains its minimum value.

Proof. Let $\alpha \in \mathbb{N}_0^s$ such that $|\alpha| = n + 1$. Since

$$m_\alpha(\cdot + i) = \sum_{\beta \leq \alpha} m_\beta(i) m_{\alpha-\beta},$$

by (1) we have

$$\begin{aligned} Qm_\alpha &= \sum_{i \in \mathbb{Z}^s} \left(\sum_{\beta \leq \alpha} m_\beta(i) \lambda m_{\alpha-\beta} \right) \phi(\cdot - i) \\ &= \sum_{i \in \mathbb{Z}^s} \left(\sum_{0 < \beta \leq \alpha} m_\beta(i) \lambda m_{\alpha-\beta} + \lambda m_\alpha \right) \phi(\cdot - i) \\ &= \lambda m_\alpha + \sum_{i \in \mathbb{Z}^s} \left(\sum_{0 < \beta \leq \alpha} m_\beta(i) \lambda m_{\alpha-\beta} \right) \phi(\cdot - i). \end{aligned}$$

Thus, by (6) we get

$$\begin{aligned} Qm_\alpha &= \lambda m_\alpha + \sum_{i \in \mathbb{Z}^s} \left(\sum_{0 < \beta \leq \alpha} m_\beta(i) g_{\alpha-\beta}(0) \right) \phi(\cdot - i) \\ &= \lambda m_\alpha - g_\alpha(0) + \sum_{i \in \mathbb{Z}^s} \left(\sum_{\beta \leq \alpha} m_\beta(i) g_{\alpha-\beta}(0) \right) \phi(\cdot - i). \end{aligned}$$

Taking into account that (cf. [2, p. 120])

$$g_\alpha = \sum_{\beta \leq \alpha} g_{\alpha-\beta}(0) m_\beta,$$

by (8) we have

$$Qm_\alpha = \lambda m_\alpha - g_\alpha(0) + Sg_\alpha.$$

Therefore, by (10) we can write

$$Qm_\alpha - m_\alpha = \lambda m_\alpha - g_\alpha(0) - G_\alpha.$$

Any sequence c for which $\|Qm_\alpha - m_\alpha\|_{\infty, [0,1]^s}$, $|\alpha| = n + 1$ is a minimum provides a solution of Problem 2, but that maximum on $[0, 1]^s$ achieves its minimum value if and only if $\lambda m_\alpha - g_\alpha(0)$ is the best uniform approximation by constant functions to G_α , which is the average value of the maximum and minimum values of G_α on $[0, 1]^s$ (cf. [24]). This completes the proof. $\square$

Remark 4. Note that the construction we propose depends on the splines G_α which only depend on ϕ .

4. C^2 quartic bivariate examples

The construction we have done is a general method for constructing discrete, differential and integral quasi-interpolants from a given s -variate B-spline. Its drawback lies in the difficulty to compute the exact values $\max_{[0,1]^s} G_\alpha$ and $\min_{[0,1]^s} G_\alpha$

for all $|\alpha| = n + 1$ when $s \geq 3$. Thus, in this section we consider some useful box splines in practice. More specifically, we take as ϕ the centered bivariate box spline associated with one of the direction sets

$$\mathcal{E}_1 = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}, \quad \mathcal{E}_2 = \begin{pmatrix} 1 & 0 & 1 & 1 & -1 & -1 \\ 0 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}, \quad \mathcal{E}_3 = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}, \quad (11)$$

that will be denoted ϕ_1 , ϕ_2 , and ϕ_3 , respectively (see e.g. [1, p. 10], [2, p. 17], [8, chapter 12]). The box spline ϕ_1 (resp. ϕ_2 and ϕ_3) is a C^2 quartic piecewise polynomial function on the uniform mesh τ_1 (resp. τ_2) generated by the directions $d_1 := (1, 0)$, $d_2 := (0, 1)$, $d_3 := d_1 + d_2$ (resp. d_1 , d_2 , d_3 , and $d_4 := -d_1 + d_2$). It is well known (see e.g. [2, p. 24]) that $\mathbb{P}_3$ is the space of maximal total degree included in $\mathcal{S}(\phi)$, so we minimize the quasi-interpolation errors for the quartic monomials. After characterizing the solutions of Problem 2, we obtain differential and integral operators. The discrete case has been considered in [17] for ϕ_r and the Zwart–Powell element (cf. [25,26]) by computing in a different way the splines from which the quasi-interpolation errors for the quartic monomials are minimized.

4.1. Characterizing the solutions of the minimization problem

First of all, the nonzero values of $\phi = \phi_r$ at the integers are needed to compute the polynomials g_α in (7) and to determine the exactness conditions (6). Moreover, the polynomials g_α enter into the construction of the splines G_α in (10), and they are required to determine the additional conditions (9). Since

$$\begin{aligned} \phi_1(0, 0) &= \frac{1}{2}, & \phi_1(1, 0) &= \phi_1(1, 1) = \phi_1(0, 1) = \phi_1(-1, 0) = \phi_1(-1, -1) = \phi_1(0, -1) = \frac{1}{12}, \\ \phi_2(0, 0) &= \frac{11}{32}, & \phi_2(1, 0) &= \phi_2(0, 1) = \phi_2(-1, 0) = \phi_2(0, -1) = \frac{1}{8}, \\ \phi_2(2, 0) &= \phi_2(0, 2) = \phi_2(-2, 0) = \phi_2(0, -2) = \frac{1}{384}, \\ \phi_2(1, 1) &= \phi_2(-1, 1) = \phi_2(-1, -1) = \phi_2(1, -1) = \frac{7}{192}, \\ \phi_3(0, 0) &= \frac{5}{12}, & \phi_3(1, 0) &= \phi_3(0, 1) = \phi_3(-1, 0) = \phi_3(0, -1) = \frac{1}{8}, \\ \phi_3(1, 1) &= \phi_3(-1, 1) = \phi_3(-1, -1) = \phi_3(1, -1) = \frac{1}{48}, \end{aligned}$$

we get

$$\begin{aligned} g_{0,0} &= 1, \\ g_{1,0} &= m_{1,0}, & g_{0,1} &= m_{0,1}, \\ g_{2,0} &= m_{2,0} - A, & g_{1,1} &= m_{1,1} - B, & g_{0,2} &= m_{0,2} - A, \\ g_{3,0} &= m_{3,0} - Am_{1,0}, & g_{2,1} &= m_{2,1} - Bm_{1,0} - Am_{0,1}, \\ g_{1,2} &= m_{1,2} - Am_{1,0} - Bm_{0,1}, & g_{0,3} &= m_{0,3} - Am_{0,1}, \\ g_{4,0} &= m_{4,0} - Am_{2,0} + C, & g_{3,1} &= m_{3,1} - Bm_{2,0} - Am_{1,1} + D, \\ g_{2,2} &= m_{2,2} - Am_{2,0} - Bm_{1,1} - Am_{0,2} + E, \\ g_{1,3} &= m_{1,3} - Am_{1,1} - Bm_{0,2} + D, & g_{0,4} &= m_{0,4} - Am_{0,2} + C, \end{aligned}$$

with

$$(A, B, C, D, E) := \begin{cases} \left(\frac{1}{6}, \frac{1}{6}, \frac{1}{72}, \frac{1}{36}, \frac{1}{24} \right) & \text{if } \phi = \phi_1, \\ \left(\frac{5}{24}, 0, \frac{3}{128}, 0, \frac{29}{576} \right) & \text{if } \phi = \phi_2, \\ \left(\frac{1}{6}, 0, \frac{1}{72}, 0, \frac{5}{144} \right) & \text{if } \phi = \phi_3. \end{cases}$$

After some computations, we determine the maximum and minimum values of G_α on $[0, 1]^2$ for $\alpha \in \{(4, 0), (3, 1), (2, 2)\}$. They appear in Table 1. The values for $G_{1,3}$ and $G_{0,4}$ follow by symmetry. Note that $G_{4,0} = G_{0,4} \equiv 0$ when $\phi = \phi_2$ and $G_{3,1} = G_{1,3} \equiv 0$ when $\phi = \phi_3$.

Thus, we obtain the following equations that characterize the solutions of Problem 2:

$$\begin{aligned} \lambda m_{0,0} &= 1, & \lambda m_{1,0} &= \lambda m_{0,1} = 0, \\ \lambda m_{2,0} &= \lambda m_{0,2} = \gamma_1, & \lambda m_{1,1} &= \gamma_2, & \lambda m_{3,0} &= \lambda m_{2,1} = \lambda m_{1,2} = \lambda m_{0,3} = 0, \\ \lambda m_{4,0} &= \lambda m_{0,4} = \gamma_3, & \lambda m_{3,1} &= \lambda m_{1,3} = \gamma_4, & \lambda m_{2,2} &= \gamma_5, \end{aligned} \quad (12)$$

Table 1Maximum and minimum values of G_α on $[0, 1]^2$.

	ϕ_1		ϕ_2		ϕ_3	
	max	min	max	min	max	min
$G_{4,0}$	$\frac{1}{384}$	0	0	0	$\frac{1}{384}$	0
$G_{3,1}$	$\frac{1}{192}$	0	$\frac{1}{1536}$	$-\frac{1}{1536}$	0	0
$G_{2,2}$	$\frac{1}{162}$	0	$\frac{1}{384}$	0	0	$-\frac{1}{192}$

where

$$(\gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5) := \begin{cases} \left(-\frac{1}{6}, -\frac{1}{6}, \frac{35}{2304}, \frac{35}{1152}, \frac{29}{648}\right) & \text{if } \phi = \phi_1, \\ \left(-\frac{5}{24}, 0, \frac{3}{128}, 0, \frac{119}{2304}\right) & \text{if } \phi = \phi_2, \\ \left(-\frac{1}{6}, 0, \frac{35}{2304}, 0, \frac{37}{1152}\right) & \text{if } \phi = \phi_3. \end{cases}$$

Next, we particularize the construction to the differential and integral cases.

4.2. C^2 quartic differential quasi-interpolants

Eq. (12) provide a linear system on the sequence c that defines the linear functional λ in (3). It depends on a given value $\ell \in \mathbb{N}$ and specific subsets J_α , $|\alpha| \leq \ell$. We choose $\ell = 2$, $J_{1,0} = J_{0,1} = J_{1,1} = \emptyset$, and $J_{0,0}, J_{2,0}, J_{1,1}, J_{0,2}$ depending on the box spline ϕ .

- If $\phi = \phi_1$, we set

$$J_{0,0} = J_{2,0} = J_{0,2} = \{(0, 0), (\pm 1, 0), (0, \pm 1), (1, 1), (-1, -1)\}.$$

Moreover, we impose some quite natural conditions on the coefficients $\beta_j := c_{(2,0),j}$, $j \in J_{2,0}$ and $\gamma_j := c_{(0,2),j}$, $j \in J_{0,2}$ defining the linear functional λ given by (3):

$$\beta_{1,0} = \beta_{0,1} = \beta_{-1,0} = \beta_{0,-1}, \quad \beta_{-1,-1} = \beta_{1,1}, \quad \gamma_{-1,0} = \beta_{0,1}, \quad \gamma_{0,-1} = \beta_{-1,0}, \quad \gamma_{-1,-1} = \beta_{1,1}.$$

No condition is imposed on $\alpha_j := c_{(0,0),j}$, $j \in J_{0,0}$. Then, system (12) has the following unique solution

$$\begin{aligned} \alpha_{0,0} &= \frac{941}{432}, & \alpha_{1,0} = \alpha_{0,1} = \alpha_{-1,0} = \alpha_{0,-1} &= -\frac{437}{1728}, & \alpha_{1,1} = \alpha_{-1,-1} &= -\frac{1}{12}, \\ \beta_{0,0} &= \frac{569}{10368}, & \beta_{1,0} = \beta_{0,1} = \beta_{-1,0} = \beta_{0,-1} &= \frac{293}{20736}, & \beta_{1,1} = \beta_{-1,-1} &= \frac{67}{2304}, \\ \gamma_j &= \beta_j, \quad j \in J_{0,2}. \end{aligned}$$

The corresponding quasi-interpolation operator Q_1^* can be expressed as

$$Q_1^* f = \sum_{i \in \mathbb{Z}^2} (f(i) L_{1,(0,0)}^* (\cdot - i) + \partial^{(2,0)} f(i) L_{1,(2,0)}^* (\cdot - i) + \partial^{(0,2)} f(i) L_{1,(0,2)}^* (\cdot - i)), \quad (13)$$

where

$$L_{1,(0,0)}^* = \frac{941}{432} \phi_1 - \frac{437}{1728} (\phi_1(\cdot \pm d_1) + \phi_1(\cdot \pm d_2)) - \frac{1}{12} \phi_1(\cdot \pm d_3), \quad (14)$$

$$L_{1,(2,0)}^* = L_{1,(2,0)}^* = \frac{569}{10368} \phi_1 + \frac{293}{20736} (\phi_1(\cdot \pm d_1) + \phi_1(\cdot \pm d_2)) + \frac{67}{2304} \phi_1(\cdot \pm d_3). \quad (15)$$

- If $\phi = \phi_2$, we choose

$$J_{0,0} = J_{2,0} = J_{0,2} = \{(0, 0), (\pm 1, 0), (0, \pm 1)\}.$$

The additional conditions

$$\alpha_{1,0} = \alpha_{-1,0}, \alpha_{0,1} = \alpha_{0,-1}, \beta_{1,0} = \gamma_{-1,0} = 0$$

yield a system (12) having the unique solution

$$\alpha_{0,0} = -\frac{1}{8}, \quad \alpha_{1,0} = \alpha_{0,1} = \alpha_{-1,0} = \alpha_{0,-1} = \frac{9}{32},$$

$$\beta_{0,0} = -\frac{683}{1152}, \quad \beta_{1,0} = \beta_{0,1} = 0, \quad \beta_{-1,0} = \beta_{0,-1} = \frac{119}{2304}$$

$$\gamma_{0,0} = -\frac{47}{96}, \quad \gamma_{1,0} = \gamma_{0,1} = \gamma_{-1,0} = \gamma_{0,-1} = 0.$$

Therefore, we obtain the quasi-interpolation operator Q_2^* given by

$$Q_2^* f = \sum_{i \in \mathbb{Z}^2} \left(f(i) L_{2,(0,0)}^* (\cdot - i) + \partial^{(2,0)} f(i) L_{2,(2,0)}^* (\cdot - i) + \partial^{(0,2)} f(i) L_{2,(0,2)}^* (\cdot - i) \right), \quad (16)$$

with

$$L_{2,(0,0)}^* = -\frac{1}{8} \phi_2 + \frac{9}{32} (\phi_2 (\cdot \pm d_1) + \phi_2 (\cdot \pm d_2)), \quad (17)$$

$$L_{2,(2,0)}^* = -\frac{683}{1152} \phi_2 + \frac{119}{2304} \phi_2 (\cdot \pm d_1), \quad L_{2,(0,2)}^* = -\frac{47}{96} \phi_2. \quad (18)$$

Note that the conditions on $\beta_{1,0}$ and $\gamma_{-1,0}$ provide fundamental functions $L_{2,(2,0)}^*$ and $L_{2,(0,2)}^*$ with small supports.

- Finally, we select the same subsets and additional conditions as above when $\phi = \phi_3$. Once again, we get a linear system (12) with the following unique solution:

$$\alpha_{0,0} = \frac{13}{48}, \quad \alpha_{1,0} = \alpha_{0,1} = \alpha_{-1,0} = \alpha_{0,-1} = \frac{35}{192},$$

$$\beta_{0,0} = -\frac{119}{288}, \quad \beta_{1,0} = \beta_{0,1} = 0, \quad \beta_{-1,0} = \beta_{0,-1} = \frac{37}{1152}$$

$$\gamma_{0,0} = -\frac{67}{192}, \quad \gamma_{1,0} = \gamma_{0,1} = \gamma_{-1,0} = \gamma_{0,-1} = 0.$$

The associated operator Q_3^* is given by

$$Q_3^* f = \sum_{i \in \mathbb{Z}^2} \left(f(i) L_{3,(0,0)}^* (\cdot - i) + \partial^{(2,0)} f(i) L_{3,(2,0)}^* (\cdot - i) + \partial^{(0,2)} f(i) L_{3,(0,2)}^* (\cdot - i) \right), \quad (19)$$

with

$$L_{3,(0,0)}^* = \frac{13}{48} \phi_3 + \frac{35}{192} (\phi_3 (\cdot \pm d_1) + \phi_3 (\cdot \pm d_2)), \quad (20)$$

$$L_{3,(2,0)}^* = -\frac{119}{288} \phi_3 + \frac{37}{1152} \phi_3 (\cdot \pm d_1), \quad L_{3,(0,2)}^* = -\frac{67}{192} \phi_3. \quad (21)$$

4.3. C^2 quartic integral quasi-interpolants

The approach we have presented for defining quasi-interpolants starts by characterizing the solutions to the minimization Problem 2. Then, after choosing the linear functional, we form the linear system (12), and we solve it. In Section 4.1, we have characterized such solutions when ϕ is one of the box splines ϕ_r , and then we have focused our attention on the construction of differential quasi-interpolants, adding some additional constraints. Now, we use (12) for deriving integral quasi-interpolants when ϕ is one of the box splines ϕ_r . By (4), the linear functional λ is given by the expression $\lambda f = \sum_{j \in J} c_j \langle f, \psi(\cdot - j) \rangle$, where ψ is another B-spline and J a finite subset of $\mathbb{Z}^2$.

Since

$$\lambda m_\alpha = \sum_{\beta \leq \alpha} \frac{1}{\beta!} \mu_\beta(\psi) \Gamma_{\alpha-\beta}(c),$$

where $\Gamma_\nu(c) := \sum_{j \in J} c_j m_\nu(j)$, $\nu \geq 0$, and $\mu_\beta(\psi) := \int_{\mathbb{R}^2} x^\beta \psi(x) dx$ is the moment of ψ of order $\beta \geq 0$, we deduce that (12) is a linear system on $(c_j)_{j \in J}$ whose matrix of coefficients involves the moments of ψ . They can be computed from the Taylor expansion of the Fourier transform of ψ :

$$\widehat{\psi}(u) = \sum_{\nu \geq 0} \partial^\nu \widehat{\psi}(0) \frac{u^\nu}{\nu!} = \sum_{\nu \geq 0} (-i)^{|\nu|} \mu_\nu(\psi) \frac{u^\nu}{\nu!}.$$

When $\psi = \phi_r$, the nonzero moments up to the fourth order are the following:

$$\mu_{0,0}(\phi_1) = 1, \quad \mu_{2,0}(\phi_1) = \mu_{0,2}(\phi_1) = \frac{1}{3}, \quad \mu_{1,1}(\phi_1) = \frac{1}{6},$$

$$\begin{array}{cccccc}
c_{0,2} & c_{1,2} & c_{2,2} & & d & e & d & a = \frac{117307}{51840} \\
c_{-1,1} & c_{0,1} & c_{1,1} & c_{2,1} & c & b & c & e & b = -\frac{17389}{103680} \\
c_{-2,0} & c_{-1,0} & c_{0,0} & c_{1,0} & c_{2,0} & d & b & a & b & d & c = -\frac{4411}{25920} \\
c_{-2,-1} & c_{-1,-1} & c_{0,-1} & c_{1,-1} & & e & c & b & c & e & d = \frac{1471}{11520} \\
c_{-2,-1} & c_{-1,-2} & c_{0,-2} & & & d & e & d & & & e = -\frac{1951}{11520}
\end{array}$$

Fig. 1. Sequence c defining the linear functional of the integral quasi-interpolant associated with the box spline ϕ_1 for the index set $J = [(\pm 2, 0), (0, \pm 2), (\pm 2, \pm 2)] \cap \mathbb{Z}^2$ after imposing some additional constraints.

$$\begin{array}{cccccc}
c_{0,2} & & a = \frac{1957}{576} & & d & & a = \frac{11071}{2880} \\
c_{-1,1} & c_{0,1} & c_{1,1} & & b = -\frac{947}{1152} & c & b & c & b = -\frac{11}{12} \\
c_{-2,0} & c_{-1,0} & c_{0,0} & c_{1,0} & c_{2,0} & c = \frac{319}{2304} & d & b & a & b & d & c = \frac{689}{5760} \\
c_{-1,-1} & c_{0,-1} & c_{1,-1} & & d = \frac{97}{1152} & c & b & c & d = \frac{991}{11520} \\
c_{0,-2} & & \phi_2 & & d & & \phi_3
\end{array}$$

Fig. 2. Sequences c defining the linear functionals of the integral quasi-interpolants associated with ϕ_2 (left) and ϕ_3 for the index set $J = [(\pm 2, 0), (0, \pm 2)] \cap \mathbb{Z}^2$.

$$\begin{aligned}
\mu_{4,0}(\phi_1) &= \mu_{0,4}(\phi_1) = \frac{3}{10}, & \mu_{3,1}(\phi_1) &= \mu_{2,2}(\phi_1) = \mu_{1,3}(\phi_1) = \frac{3}{20}, \\
\mu_{0,0}(\phi_2) &= 1, & \mu_{2,0}(\phi_2) &= \mu_{0,2}(\phi_2) = \frac{5}{12}, \\
\mu_{4,0}(\phi_2) &= \mu_{0,4}(\phi_2) = \frac{23}{48}, & \mu_{2,2}(\phi_2) &= \frac{101}{720}, \\
\mu_{0,0}(\phi_3) &= 1, & \mu_{2,0}(\phi_3) &= \mu_{0,2}(\phi_3) = \frac{1}{3}, \\
\mu_{4,0}(\phi_3) &= \mu_{0,4}(\phi_3) = \frac{3}{10}, & \mu_{2,2}(\phi_3) &= \frac{17}{180}.
\end{aligned}$$

- If $\phi = \phi_1$, we set $J = [(\pm 2, 0), (0, \pm 2), (\pm 2, \pm 2)] \cap \mathbb{Z}^2$. For this choice, the associated linear system (12) has infinitely many solutions. We can impose additional conditions in order to find a particular solution having some interesting properties in practice. For instance, we are interested in hexagonal solutions, i.e. sequences c such that $c_{G_i\alpha} = c_\alpha$ for all $\alpha \in J$ and $i = 1, 2$, with

$$G_1 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \quad \text{and} \quad G_2 = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}.$$

However, there is no hexagonal solution. Instead, we impose some of these conditions, namely $c_{1,0} = c_{-1,0}$, $c_{0,1} = c_{0,-1}$, $c_{1,1} = c_{-1,-1}$ and $c_{1,1} = c_{1,-1}$, and we get the solution shown in Fig. 1.

- For $\phi = \phi_2$ or ϕ_3 , we set $J = [(\pm 2, 0), (0, \pm 2)] \cap \mathbb{Z}^2$. Each one of the corresponding linear systems (12) has a unique solution. Both sequences appear in Fig. 2. They are Lozenge sequences, i.e. $c_{G_i\alpha} = c_\alpha$ for all $\alpha \in J$ and $i = 1, 2$, with

$$G_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad G_2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

5. Error estimates and numerical tests for the differential quasi-interpolants

The next result provides error estimates in the max-norm for the scaled differential operators associated with Q_1^* , Q_2^* , and Q_3^* on each triangle T in the triangulation $h\tau$ obtained by shrinking τ by a factor $h > 0$. We will use the notation

$$\|D^r u\|_{\infty,S} := \sup_{x \in S} \sup \{ |D^r u(x)(v_1, \dots, v_r)| : v_i \in \mathbb{R}^2, \|v_i\| = 1, 1 \leq i \leq r \}$$

for the uniform norm on the compact set S of the r th order total derivative $D^r u$ of u , $\|v\|$ being the usual Euclidean norm in $\mathbb{R}^2$ for v .

Proposition 5. Let $T \in h\tau$, $h > 0$. For each $1 \leq r \leq 3$, there are constants $K_{r,\alpha}$ independent of f , h and T , and a neighbourhood V of T such that

$$\|\partial^\alpha (Q_{r,h}^* f - f)\|_{\infty,T} \leq K_{r,\alpha} h^{4-|\alpha|} \|D^4 f\|_{\infty,V}, \quad |\alpha| \leq 2.$$

Moreover,

$$\begin{array}{lll} K_{1,\{0,0\}}^* \approx 0.859, & K_{2,\{0,0\}}^* \approx 0.894, & K_{3,\{0,0\}}^* \approx 0.563, \\ K_{1,\{1,0\}}^* \approx 1.995, & K_{2,\{1,0\}}^* \approx 1.715, & K_{3,\{1,0\}}^* \approx 0.992, \\ K_{1,\{0,1\}}^* \approx 1.995, & K_{2,\{0,1\}}^* \approx 1.790, & K_{3,\{0,1\}}^* \approx 1.094, \\ K_{1,\{2,0\}}^* \approx 3.613, & K_{2,\{2,0\}}^* \approx 3.229, & K_{3,\{2,0\}}^* \approx 3.006, \\ K_{1,\{1,1\}}^* \approx 4.801, & K_{2,\{1,1\}}^* \approx 3.046, & K_{3,\{1,1\}}^* \approx 1.973, \\ K_{1,\{0,2\}}^* \approx 3.613, & K_{2,\{0,2\}}^* \approx 3.280, & K_{3,\{0,2\}}^* \approx 3.192. \end{array}$$

Proof. The proof we give here imitates the approach already given in [17] for the discrete case. By (5), we have

$$Q_{r,h}^* f = \sum_{i \in \mathbb{Z}^2} f(hi) L_{r,\{0,0\}} \left(\frac{\cdot}{h} - i \right) + h^2 \sum_{i \in \mathbb{Z}^2} \partial^{(2,0)} f(hi) L_{r,\{2,0\}} \left(\frac{\cdot}{h} - i \right) + h^2 \sum_{i \in \mathbb{Z}^2} \partial^{(0,2)} f(hi) L_{r,\{0,2\}} \left(\frac{\cdot}{h} - i \right),$$

where the functions $L_{r,\{0,0\}}$, $L_{r,\{2,0\}}$, and $L_{r,\{0,2\}}$ are given by (14)–(15), (17)–(18), or (20)–(21).

If $\phi = \phi_1$ (resp. ϕ_2 or ϕ_3), for a given $x \in T$, x lies in one of the three (resp. four) triangles T_p obtained by drawing the diagonals of the square with opposite vertices ah and $(a + d_3)h$ (resp. $(a - \frac{1}{2}d_3)h$ and $(a + \frac{1}{2}d_3)h$, or ah and $(a + d_3)h$) for some $a \in \mathbb{Z}^2$. The barycenter of T_p is $\eta_p = (a + \omega_p)h$, where

$$\begin{cases} \omega_1 = \left(\frac{2}{3}, \frac{1}{3} \right), & \omega_2 = \left(\frac{1}{3}, \frac{2}{3} \right) & \text{if } \phi = \phi_1, \\ \omega_1 = \left(\frac{1}{3}, 0 \right), & \omega_2 = \left(0, \frac{1}{3} \right), & \omega_3 = \left(-\frac{1}{3}, 0 \right), & \omega_4 = \left(0, -\frac{1}{3} \right) & \text{if } \phi = \phi_2, \\ \omega_1 = \left(\frac{5}{6}, \frac{1}{2} \right), & \omega_2 = \left(\frac{1}{2}, \frac{5}{6} \right), & \omega_3 = \left(\frac{1}{6}, \frac{1}{2} \right), & \omega_4 = \left(\frac{1}{2}, \frac{1}{6} \right) & \text{if } \phi = \phi_3. \end{cases}$$

The Taylor formulae

$$\begin{aligned} f(x) &= f(\eta_p) + Df(\eta_p)(x - \eta_p) + \frac{1}{2} D^2 f(\eta_p)(x - \eta_p)^2 + \frac{1}{6} D^3 f(\eta_p)(x - \eta_p)^3 + R_0(x), \\ R_0(x) &= \frac{1}{6} \int_0^1 (1-t)^3 D^4 f((1-t)\eta_p + tx)(x - \eta_p)^4 dt, \\ D^2 f(x)(w_1, w_2) &= D^2 f(\eta_p)(w_1, w_2) + Df(\eta_p)(x - \eta_p, w_1, w_2) + R_2(x, w_1, w_2), \\ R_0(x, w_1, w_2) &= \int_0^1 (1-t) D^4 f((1-t)\eta_p + tx)(x - \eta_p, x - \eta_p, w_1, w_2) dt, \end{aligned}$$

where $w_1, w_2 \in \mathbb{R}^2 \setminus \{0\}$, allow us to expand each term $f(hi)$, $\partial^{(2,0)} f(hi)$ and $\partial^{(0,2)} f(hi)$ in the equality for $Q_{r,h}^* f$. Taking into account the exactness of $Q_{r,h}$ on $\mathbb{P}_3$, after some algebra we get

$$\begin{aligned} Q_{r,h}^* f(x) - f(x) &= \sum_{i \in \mathbb{Z}^2} R_0(ih) L_{r,\{0,0\}} \left(\frac{x}{h} - i \right) + h^2 \sum_{i \in \mathbb{Z}^2} R_2(ih, d_1, d_1) L_{r,\{2,0\}} \left(\frac{x}{h} - i \right) \\ &\quad + h^2 \sum_{i \in \mathbb{Z}^2} R_2(ih, d_2, d_2) L_{r,\{0,2\}} \left(\frac{x}{h} - i \right) - R_0(x). \end{aligned}$$

If $x \in T_p$, then $\xi_p := \frac{x}{h} - a$ belongs to the triangle $\Delta_p \in \tau$ with barycenter ω_p . Moreover,

$$\begin{aligned} Q_{r,h}^* f(x) - f(x) &= \sum_{i \in \mathbb{Z}^2} R_0((a+i)h) L_{r,\{0,0\}}(\xi_p) + h^2 \sum_{i \in \mathbb{Z}^2} R_2((a+i)h, d_1, d_1) L_{r,\{2,0\}}(\xi_p) \\ &\quad + h^2 \sum_{i \in \mathbb{Z}^2} R_2((a+i)h, d_2, d_2) L_{r,\{0,2\}}(\xi_p) - R_0(x). \end{aligned}$$

Let $V := h(a + [I_p])$, $[I_p]$ denoting the convex hull of the subset I_p formed by the integers i such that the support of $L_{0,0}(\cdot - i)$, $L_{2,0}(\cdot - i)$ or $L_{0,2}(\cdot - i)$ intersects the interior of Δ_p . Since

$$|R_0((a+i)h)| \leq \frac{h^4}{24} \|i - \omega_p\|^4 \|D^4 f\|_{\infty, V},$$

and

$$|R_2((a+i)h, d_j, d_j)| \leq \frac{h^2}{2} \|i - \omega_p\|^2 \|D^4 f\|_{\infty, V}, \quad j = 1, 2,$$

we get

$$|Q_{r,h}^* f(x) - f(x)| \leq h^4 \Lambda(\xi_p) \|D^4 f\|_{\infty, V},$$

where

$$\begin{aligned} \Lambda(\xi_p) &:= \frac{1}{24} \|\xi_p - \omega_p\|^4 + \frac{1}{24} \sum_{i \in I_p} \|i - \omega_p\|^4 |L_{r,\{0,0\}}(\xi_p - i)| \\ &\quad + \frac{1}{2} \sum_{i \in I_p} \|i - \omega_p\|^2 |L_{r,\{2,0\}}(\xi_p - i)| + \frac{1}{2} \sum_{i \in I_p} \|i - \omega_p\|^2 |L_{r,\{0,2\}}(\xi_p - i)|. \end{aligned}$$

Therefore, the claim follows with

$$K_{r,\{0,0\}}^* = \max_p \max_{\xi_p \in \Delta_p} \Lambda(\xi_p).$$

For the first order partial derivatives of the quasi-interpolation error, we start from the expansion

$$Df(x)w = Df(\eta_p)w + D^2 f(\eta_p)(x - \eta_p, w) + \frac{1}{2} D^3 f(\eta_p)(x - \eta_p, w, w) + R_1(x, w),$$

where

$$R_1(x, w) = \frac{1}{2} \int_0^1 (1-t)^2 D^4 f((1-t)\eta_p + tx) \left((x - \eta_p)^3, w \right) dt.$$

A computation as above provides for $x \in T_p$ and w such that $\|w\| = 1$ the estimate

$$|D(Q_{r,h}^* f - f)(x)w| \leq h^3 \Lambda_w(\xi_p) \|D^4 f\|_{\infty, V},$$

with

$$\begin{aligned} \Lambda_w(\xi_p) &:= \frac{1}{6} \|\xi_p - \omega_p\|^3 + \frac{1}{24} \sum_{i \in I_p} \|i - \omega_p\|^4 |DL_{r,\{0,0\}}(\xi_p - i)w| \\ &\quad + \frac{1}{2} \sum_{i \in I_p} \|i - \omega_p\|^2 |DL_{r,\{2,0\}}(\xi_p - i)w| + \frac{1}{2} \sum_{i \in I_p} \|i - \omega_p\|^2 |DL_{r,\{0,2\}}(\xi_p - i)w|. \end{aligned}$$

Thus, we take $K_{r,(1,0)}^* = \max_p \max_{\xi_p \in \Delta_p} \Lambda_{d_1}(\xi_p)$, and $K_{r,(0,1)}^* = \max_p \max_{\xi_p \in \Delta_p} \Lambda_{d_2}(\xi_p)$. Similarly, we obtain for $x \in T_p$ and w_1 and w_2 such that $\|w_1\| = \|w_2\| = 1$ the estimate

$$|D^2(Q_{r,h}^* f - f)(x)(w_1, w_2)| \leq h^2 \Lambda_{w_1, w_2}(\xi_p) \|D^4 f\|_{\infty, V},$$

with

$$\begin{aligned} \Lambda_{w_1, w_2}(\xi_p) &= \frac{1}{2} \|\xi_p - \omega_p\|^2 + \frac{1}{24} \sum_{i \in I_p} \|i - \omega_p\|^4 |D^2 L_{r,\{0,0\}}(\xi_p - i)(w_1, w_2)| \\ &\quad + \frac{1}{2} \sum_{i \in I_p} \|i - \omega_p\|^2 |D^2 L_{r,\{2,0\}}(\xi_p - i)(w_1, w_2)| + \frac{1}{2} \sum_{i \in I_p} \|i - \omega_p\|^2 |D^2 L_{r,\{0,2\}}(\xi_p - i)(w_1, w_2)|, \end{aligned}$$

from which we get $K_{r,(2,0)}^* = \max_p \max_{\xi_p \in \Delta_p} \Lambda_{d_1, d_1}(\xi_p)$, $K_{r,(1,1)}^* = \max_p \max_{\xi_p \in \Delta_p} \Lambda_{d_1, d_2}(\xi_p)$, and $K_{r,(0,2)}^* = \max_p \max_{\xi_p \in \Delta_p} \Lambda_{d_2, d_2}(\xi_p)$. It only remains to compute the maxima on Δ_p of Λ , Λ_{d_j} , and Λ_{d_j, d_k} , $1 \leq j, k \leq 2$, to complete the proof. $\square$

The estimates in Proposition 5 are pessimistic. As we allow some oversampling, the quasi-interpolants are expressed in terms of some functions having large supports. Then, there are many indices involved in the sums defining Λ , Λ_{d_j} , and Λ_{d_j, d_k} , and so large values are expected. We compare these quasi-interpolants with some other obtained by a classical method and having fundamental functions with smaller supports.

To test the differential quasi-interpolation operators $Q_{r,h}^*$, we consider the Franke and Nielson reference functions (cf. [27,28])

$$\begin{aligned} Fr(x, y) &= 0.75 \exp\left(-\frac{1}{4}((9x-2)^2 + (9y-2)^2)\right) + 0.75 \exp\left(-\frac{1}{49}(9x+1)^2 - \frac{1}{10}(9y+1)\right) \\ &\quad + 0.5 \exp\left(-\frac{1}{4}((9x-7)^2 + (9y-3)^2)\right) - 0.2 \exp(-(9x-4)^2 - (9x-7)^2), \\ N(x, y) &= \frac{y}{2} \cos^4(4(x^2 + y - 1)), \end{aligned}$$

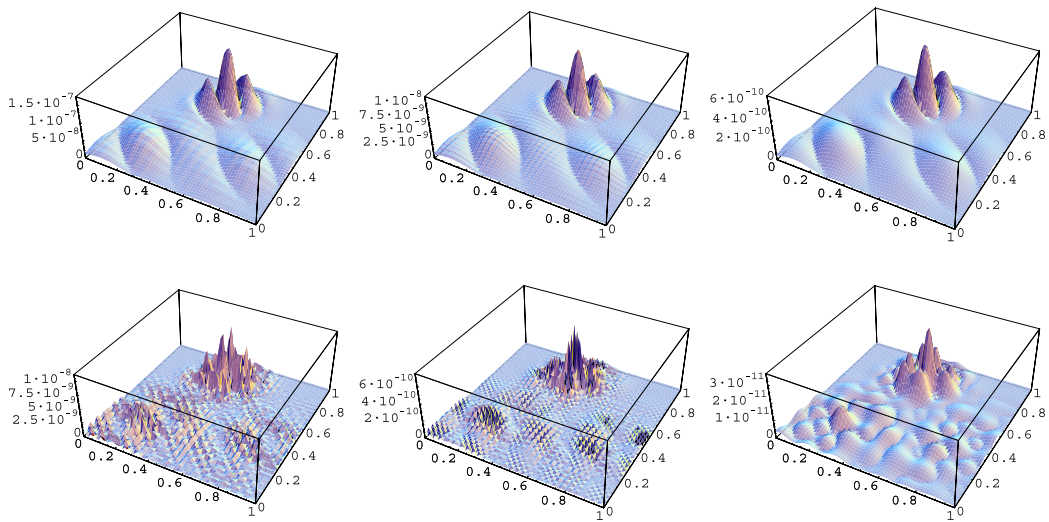


Fig. 3. From left to right, plots of $|\tilde{Q}_{1,h}Fr - Fr|$ (top) and $|Q_{1,h}^*Fr - Fr|$ with step length $h = 2^{-n}$ and $8 \leq n \leq 10$.

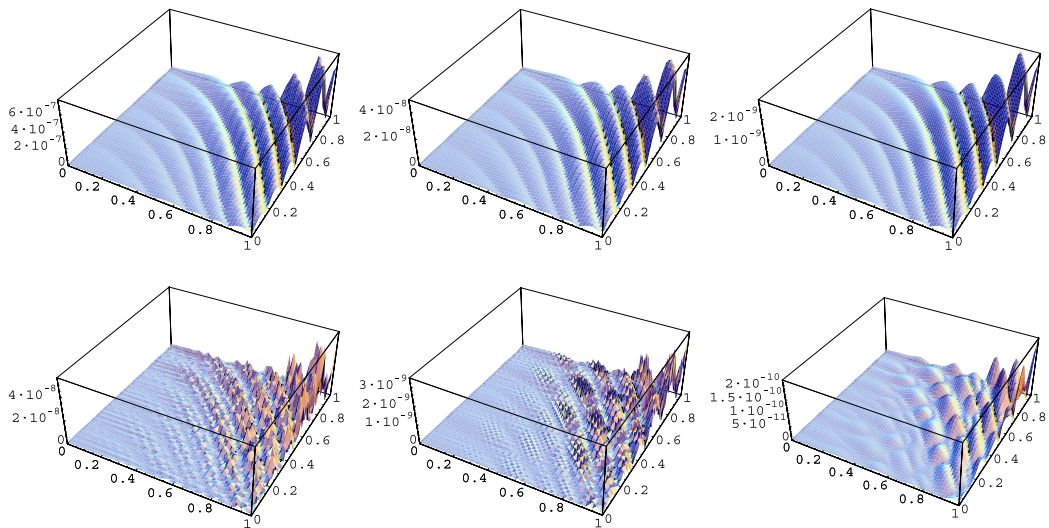


Fig. 4. From left to right, plots of $|\tilde{Q}_{1,h}N - N|$ (top) and $|Q_{1,h}^*N - N|$ with step length $h = 2^{-n}$ and $8 \leq n \leq 10$.

and the classical differential quasi-interpolants $\tilde{Q}_r f$ given by

$$\tilde{Q}_r f = \sum_{i \in \mathbb{Z}^2} \tilde{\lambda}_r f(\cdot + i) \phi_r(\cdot - i),$$

where

$$\tilde{\lambda}_1 f = f(0) - \frac{1}{6} (\partial^{(2,0)} f(0) + \partial^{(1,1)} f(0) + \partial^{(0,2)} f(0)),$$

and, for $r = 2, 3$,

$$\tilde{\lambda}_r f = f(0) - \frac{1}{6} (\partial^{(2,0)} f(0) + \partial^{(0,2)} f(0)).$$

They are obtained from the Taylor expansion of $1/\hat{\phi}$, where $\hat{\phi}$ denotes the Fourier transform of the C^2 quartic box spline ϕ (see [7]). For these quasi-interpolants we have error estimates as in Proposition 5, but now the constants are

$$\begin{aligned} \tilde{K}_{1,\{0,0\}} &\approx 0.447, & \tilde{K}_{2,\{0,0\}} &\approx 0.282, & \tilde{K}_{3,\{0,0\}} &\approx 0.229, \\ \tilde{K}_{1,\{1,0\}} &\approx 0.974, & \tilde{K}_{2,\{1,0\}} &\approx 0.590, & \tilde{K}_{3,\{1,0\}} &\approx 0.453, \\ \tilde{K}_{1,\{0,1\}} &\approx 0.974, & \tilde{K}_{2,\{0,1\}} &\approx 0.590, & \tilde{K}_{3,\{0,1\}} &\approx 0.453, \end{aligned}$$

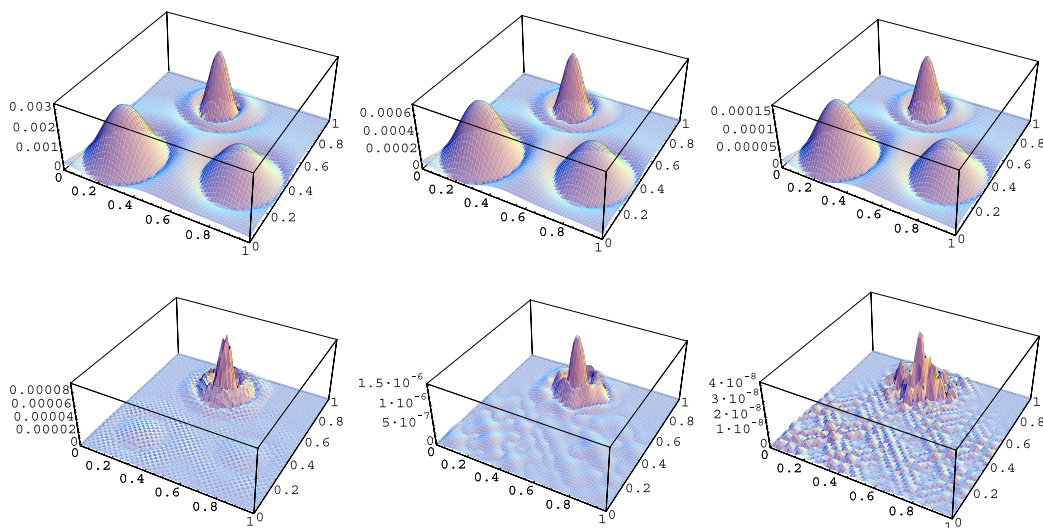


Fig. 5. From left to right, plots of $|\tilde{Q}_{2,h}Fr - Fr|$ (top) and $|Q_{2,h}^*Fr - Fr|$ with step length $h = 2^{-n}$ and $5 \leq n \leq 7$.

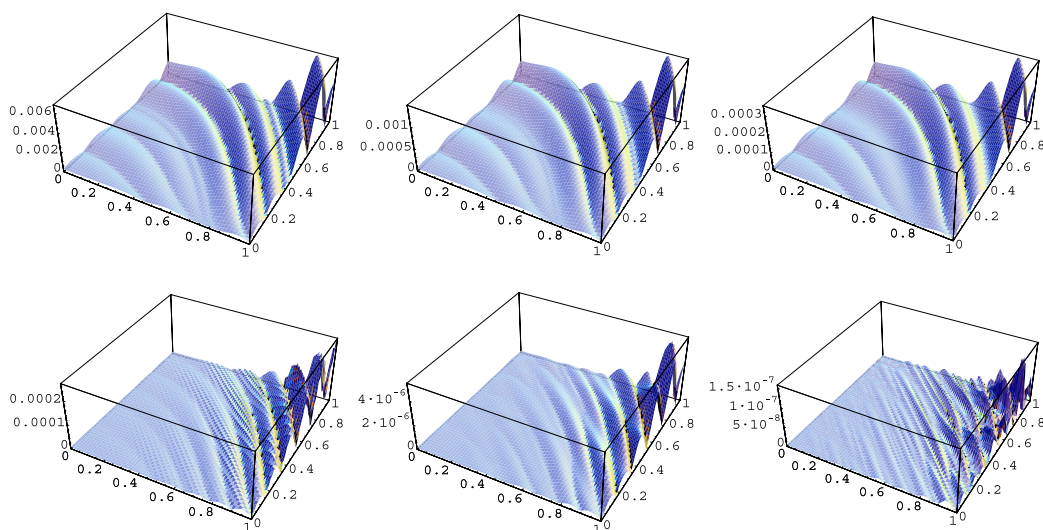


Fig. 6. From left to right, plots of $|\tilde{Q}_{2,h}F - F|$ (top) and $|Q_{2,h}^*F - F|$ with step length $h = 2^{-n}$ and $5 \leq n \leq 7$.

$$\begin{aligned} \tilde{K}_{1,\{2,0\}} &\approx 2.004, & \tilde{K}_{2,\{2,0\}} &\approx 1.185, & \tilde{K}_{3,\{2,0\}} &\approx 1.439, \\ \tilde{K}_{1,\{1,1\}} &\approx 2.579, & \tilde{K}_{2,\{1,1\}} &\approx 1.055, & \tilde{K}_{3,\{1,1\}} &\approx 0.883, \\ \tilde{K}_{1,\{0,2\}} &\approx 2.004, & \tilde{K}_{2,\{0,2\}} &\approx 1.092, & \tilde{K}_{3,\{0,2\}} &\approx 1.346. \end{aligned}$$

Fig. 3 shows the graphs of the quasi-interpolation errors $|\tilde{Q}_{1,h}Fr - Fr|$ (top line) and $|Q_{1,h}^*Fr - Fr|$ for several values of h . The corresponding graphs for the Nielson function are shown in Fig. 4. They illustrate the nearly optimal approximation order of the new quasi-interpolation schemes and the good results achieved when some oversampling is allowed.

The plots shown in Figs. 5 and 6 suggest that, in general, $Q_{2,h}^*$ produces better results than $Q_{2,h}$. Also the operator $Q_{3,h}^*$ gives better results than $\tilde{Q}_{3,h}$ for the test functions Fr and N .

6. Concluding remarks

A general method has been proposed to define discrete, differential and integral quasi-interpolants (exact on some polynomial spaces) paying attention to the quasi-interpolation error and allowing, if necessary, some oversampling. The sequences defining the resulting operators are solutions of some minimization problems and they are characterized in terms of some splines defined from the B-spline spanning the approximation space. These splines (independent of the B-spline) permit to derive additional linear equations to be added to the linear system that guarantees the required exactness. The C^2

quartic differential examples, based on the useful box splines in practice, are fully analyzed. They show that the resulting schemes are competitive when compared with the corresponding classical differential quasi-interpolation operators. Also, some integral schemes have been defined.

There is a number of possible extensions of the modified approach presented in this paper, but the most important concerns the computation of the maximum and minimum values of each spline involved in the construction. It is very complicated to compute the exact values in the s -variate case when $s \geq 3$, and then it is important to define a different objective function to be easily minimized.

Acknowledgements

We would like to thank the anonymous referees for pertinent comments which helped us to improve this paper.

References

- [1] C. de Boor, K. Höllig, S. Riemenschneider, *Box Splines*, Springer-Verlag, New York, 1993.
- [2] C.K. Chui, *Multivariate Splines*, in: CBMS-NSF Regional Conference Series in Applied Mathematics, vol. 54, SIAM, Philadelphia, 1988.
- [3] C. de Boor, The polynomials in the linear span of integer translates of a compactly supported function, *Constr. Approx.* 3 (1987) 199–208.
- [4] C.K. Chui, H. Diamond, A natural formulation of quasi-interpolation by multivariate splines, *Proc. Amer. Math. Soc.* 99 (4) (1987) 643–646.
- [5] C.K. Chui, M.J. Lai, A multivariate analog of Marsden's identity and a quasi-interpolation scheme, *Constr. Approx.* 3 (1987) 111–122.
- [6] C.K. Chui, H. Diamond, A characterization of multivariate quasi-interpolation formulas and its applications, *Numer. Math.* 57 (1990) 105–121.
- [7] W. Dahmen, C.A. Micchelli, Translates of multivariate splines, *Linear Algebra Appl.* 52/53 (1983) 217–234.
- [8] M.-J. Lai, L.L. Schumaker, *Spline Functions on Triangulations*, Cambridge University Press, Cambridge, 2007.
- [9] P. Sablonnière, Quasi-interpolants associated to H-splines on a three-direction mesh, *J. Comput. Appl. Math.* 66 (1996) 433–442.
- [10] P. Sablonnière, H-splines and quasi-interpolants on a three directional mesh, in: M.D. Buhmann, D.H. Mache (Eds.), *Advanced Problems in Constructive Approximation*, in: International Series of Numerical Mathematics, vol. 142, Birkhäuser Verlag, Basel, 2002, pp. 187–201.
- [11] D. Barrera, M.J. Ibáñez, P. Sablonnière, Near-best discrete quasi-interpolants on uniform and nonuniform partitions, in: A. Cohen, J.-L. Merrien, L.L. Schumaker (Eds.), *Curve and Surface Fitting: Saint-Malo 2002*, Nashboro Press, Brentwood, 2003, pp. 31–40.
- [12] D. Barrera, M.J. Ibáñez, P. Sablonnière, D. Sibih, Near minimally normed spline quasi-interpolants on uniform partitions, *J. Comput. Appl. Math.* 181 (2005) 211–233.
- [13] D. Barrera, M.J. Ibáñez, P. Sablonnière, D. Sibih, Near-best quasi-interpolants associated with H-splines on a three-direction mesh, *J. Comput. Appl. Math.* 183 (2005) 133–152.
- [14] D. Barrera, M.J. Ibáñez, Bernstein–Bézier representation and near-minimally normed discrete quasi-interpolation operators, *Appl. Numer. Math.* 58 (2008) 59–68.
- [15] D. Barrera, M.J. Ibáñez, P. Sablonnière, D. Sibih, On near-best discrete quasi-interpolation on a four-directional mesh, *J. Comput. Appl. Math.* 233 (2010) 1470–1477.
- [16] D. Barrera, M.J. Ibáñez, P. Sablonnière, D. Sibih, Near-best univariate spline discrete quasi-interpolants on non-uniform partitions, *Constr. Approx.* 28 (2008) 237–251.
- [17] D. Barrera, M.J. Ibáñez, Minimizing the quasi-interpolation error for bivariate discrete quasi-interpolants, *J. Comput. Appl. Math.* 224 (2009) 250–268.
- [18] D. Barrera, A. Guessab, M.J. Ibáñez, O. Nouisser, Optimal bivariate C^1 cubic quasi-interpolation on a type-2 triangulation, *Journal of Computational and Applied Mathematics* (2009), doi:10.1016/j.cam.2009.07.035.
- [19] M.J. Ibáñez Pérez, *Quasi-interpolantes spline discretos de norma casi mínima. Teoría y aplicaciones*, Doctoral Dissertation, Universidad de Granada, 2003.
- [20] M.J. Ibáñez Pérez, On Chebyshev-type discrete quasi-interpolants, *Math. Comput. Simul.* 77 (2008) 218–227.
- [21] M. Jirani, Quasi-interpolants intégraux de Chebyshev sur des partitions uniformes de la droite réelle, Technical report (November 2005), University of Oujda (Morocco).
- [22] M.A. Fortes, M.J. Ibáñez, M.L. Rodríguez, On Chebyshev-type integral quasi-interpolation operators, *Math. Comput. Simul.* 79 (2009) 3478–3491.
- [23] S. Dekel, D. Leviatan, On measuring the efficiency of kernel operators $L_p(\mathbb{R}^d)$, *Adv. Comput. Math.* 20 (2004) 53–65.
- [24] G.A. Watson, *Approximation Theory and Numerical Methods*, Wiley, Chichester, 1980.
- [25] M.J.D. Powell, Piecewise quadratic surface fitting for contour plotting, in: D.J. Evans (Ed.), *Software for Numerical Mathematics*, Academic Press, London, 1974, pp. 253–271.
- [26] P.B. Zwart, Multivariate splines with non-degenerate partitions, *SIAM J. Numer. Anal.* 10 (1973) 665–673.
- [27] R. Franke, Scattered data interpolation: Tests of some methods, *Math. Comp.* 157 (1982) 181–200.
- [28] G.M. Nielson, A first order blending method for triangles based upon cubic interpolation, *Int. J. Numer. Meth. Engng.* 15 (1978) 308–318.